BrightLight Data Analytics

Research Assignment 2

Foundations of Data & Artificial Intelligence

Course	Introduction to Data & Artificial Intelligence
Assignment No.	Research Assignment 2
Target Audience	Beginner Students
Submission	Upload on Github
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and email it to rofhiwa@brightlighttutorials.co.za before the due date.

Section A: Data Fundamentals

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Research Tip: *Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.*

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

- A database is an organised collection of information, it stores organised data so that you can quickly save, retrieve and update.
- it's essential as they are used for day-to-day tasks for storage and transactional purposes
- Examples are: Clients lists and Online banking.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

-> Data Warehouse is a centered place where we store Multiple database for analysis.
- Database is designed for transactional processing (OLTP) while Data Warehouse is optimized for analytical processing (OLAP)
- Data Warehouse are optimized for historical data storage and complex analytics. and assist in improved decision-making

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

-> Data Lake is a comprehensive cloud-based data storage and analytics service, it is designed to store/handle large volumes of data enabling organization to store and process data
- concept and difference: data warehouse requires data to be cleaned and structured before being saved (schema-on-write). A data lake allows you to store everything first and define the structure only when you need to use it (schema-on-read).
- A business would prefer a data lake when they need to store massive amounts of diverse big data for future use.

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

-> Data Mart is a simple form of a data warehouse that is focused on a single subject or functional area such as sales finance or marketing
- It is often a subset of a large data warehouse designed to make a data access easier and faster for specific group of users
-> it is typically used by specific department or organisation who needs quick access to specific data relevant to their daily operations.

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5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

- Structured Data refers to data that is organised in predefined format, typically stored in rational database with rows and column
example: customer records.
- Semi-structured data does not conform to a strict schema but still has some organised properties making it easier to analyze.
example: XML files.
- Unstructured data lacks a predefined format or organization making it more challenging to process and analyze.
example: Multimedia files (images, videos)

6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

- Data Modeling is the process of creating a simplified visual representation of a software system and the data elements it contains
-> It is important as it ensures accuracy, improve performance and has clear communications.
- Types: Logical data Model - A more detailed model that defines specific Attributes
Physical data Model - it specifies how the data will be implemented in particular DBMS.

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.

- Data Mining is a systemat process of discovering hidden patterns, correlation and meaningful insights from large databases
- uses in business: business uses data Mining to turn raw data into actionable information to improve decision making.
- examples: Retail Markets for analyzing purchases history
finance (fraud detection) to identify unusual spending patterns.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

- Data Management system is a software suite that acts as an interface between users and a database.

Platforms	Description	Suitable use case
1 MySQL	* A open source relational database know for its readability and ease of use	- Structured data like customer orders
2 PostgreSQL	* An advanced open-source relational database that emphasises extensibility	- Complex data analysis and scientific research
3 Oracle	* A powerful multi-model RDBMS designed for enterprise level Applicat	- Banking and finance.
4 MongoDB	* A popular NoSQL database that stores data in flexible JSON-like structures	- Content Management system

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9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

- ETL is a central data intergration process used to combine data from multiple sources into single, consistent data store.

Three steps:

* Extract - it involves retrieving raw data from various sources system
* Transform - Raw data is cleaned and modified to meet business needs
* Load -> it is where the processed and formated data is written into the target destination

- it is important as it provides clean, and high-quality foundation for data analysis.

10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

formats.	Description	when to use.
- CSV.	* Simple text format where data is stored in rows separated by commas	Manual data exchange or quick exports of small datasets
- JSON	* A flexible, text-based format that uses key value pairs; great for nested or semi-structured data	- APIs and web services where flexibility and human readability are important.
- Parquet	A columnar storage format optimised for big data. it uses heavy compression and is fast for reading specific colums.	- Large-scale analytics and data lakes where high performance and lower cloud costs are priorities.
- Avro	* A row-based binary format that stores the schema within the files, making it very compact.	- Streaming data and situations where structure changes frequently (schema evolution)

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Section B: Artificial Intelligence & Machine Learning

Research Tip: *Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.*

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

- Artificial intelligence is the simulation of human intelligence by computer system that helps with problem solving and learning.
- Narrow AI is designed to perform single specific task and it operates under a limited set of constraints.
  example: Apple's Siri
- General AI: The Machine has the ability to understand, learn and apply knowledge across a wide range of tasks
  example - Human Level AI

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

- Machine Learning is a subset of AI that focuses on building systems that learn from data to improve their performance on specific tasks
- In traditional Programming A Programmer writes manual rules that take input data to produce an output, while Machine Learning data and outputs are fed into an algorithm to learn the rules automatically.
- 3 Main types: Supervised Learning: The Model is trained on labeled data
  - unsupervised Learning: The Model finds hidden patterns that has no labels
  - Reinforcement Learning: An agent learns to make decisions by performing actions in an environment.

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

- Deep Learning is specialized subset of Machine learning based on artificial neural networks with many layers. It is designed to mimic the way human brains process data.
- Relation: it is a subfield of Machine learning, which is a subfield of Artificial intelligence.
- Neural Networks are computing systems inspired by the biological neural network in humans brains.
- * they work by passing data through layers, where each layer assigns weights to the data to identify features and patterns.

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

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- Natural Language Processing (NLP) is a branch of AI that enables computers to understand, interprete and generate language.
- Applications: ① Spam filters → it automatically identify and divert junk mails.
② Language Translation → like google Translate that converts one language to another
③ Voice Assistance → Tools like Siri or google Assistant that process voice commands.

15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

- Large Language Model is a type of artificial intelligence trained on vast amount of text data to understand and generate human-like language
- Difference: Standard Machine learning Models are often designed for specific narrow tasks while LLMS are ~~generated~~ general-purpose and can perform a wide variety of tasks.
* ChatGPT: used for conversational assistance, content creation and problem solving.
* Claude: used for complex reasoning, summarizing long documents and coding help.

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

- Generative AI is a branch of AI focused on creating new contents rather than just analysing existing data.
- How it works: At high level, it uses neural networks to identify patterns and structures within massive training dataset.

Tools: * DALL-E → Generates digital image from text description
* Midjourney → Creates high-quality artistic visuals and illustrations.
* Github Copilot → Suggests and completes lines of codes for programmers.

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

- Training Dataset is the initial set of data used to teach an AI model how to make predictions or perform tasks.
- Importance: High-quality data ensures the model is accurate and reliable if data is diverse and correct, the model learns the right pattern
- Consequences: if data is biased, the model will produce unfair or discriminatory results, if it is incorrect the ~~data~~ model will provide hallucination or false information making it untrustworthy.

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

- Computer vision is a field of AI that enables computers to derive meaningful information from digital images, videos and other visual inputs.

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- How Machines See: Machine interpret images by breaking them down into pixels, they use algorithms to detect edges, colors and textures; eventually recognizing complex patterns that represents objects or faces.

- Real world cases: Self-driving cars - To detect pedestrians, traffic lights and other vehicles.

* Facial Recognitions: used for securing smartphones

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Section C: APIs, Tools & Developer Concepts

Research Tip: *Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.*

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?

- An API is a set of rules and protocols that allows different software applications to communicate and share data with each other.
- real-world analogy – APIs are like the waiter in a restaurant, they take your request and send it to the kitchen to process your order.
- APIs are the "connective tissue" of modern software. They allow for integration (e.g. pulling data from multiple airlines), automation and the ability to build complex systems by leveraging existing services rather than starting from scrat

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

- An API Key is a unique identifier used to authenticate a project developer, or calling program to an API.
* Purpose: its primary purpose is to identify who is making the request and authorize access based on that identity.
* Storage and Protection: Key should never be hardcoded in public source code instead use environmental variables or secure secret management tool
- exposure consequences: if exposed, unauthorized users can use your account potentially leading to data breaches, service disruption.

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

- REST stands for Representational State Transfer. It is an architectural style for designing networked application that uses HTTP protocol to communicate
Main Methods: GET → used to retrieve a read data from a server.
* POST → used to create a new resource on the server.
* PUT → used to update or replace an existing resources entirely
* Delete → used to remove a resource from the server

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

- JSON is a lightweight, text-based format for ~~format~~ storing and exchanging data.
- it is highly popular because it is human-readable and language-independent, meaning it can be used by almost any programming language.

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* example: json
{ "name": "William",
"job_title": "Data Analyst",
"is_Active": true,
"skills": ["Python", "SQL", "Tableau"]
}

23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

- Prompt Engineering is the process of designing and refining specific inputs to guide Artificial Intelligence Models to produce high-quality and relevant output
- AI Models respond to patterns in language. A well-crafted prompt provides the necessary context, constraints and persona the AI needs to minimize hallucination and generate a response that truly meets the user's needs

Poor Prompt	Improved Prompt
* "Write a story about a dog"	- Write a 300-word heartwarming story about a golden Retriever named Max who finds a lost kitten in a rainy city. Tone: Hopeful.
* "How do I fix my code?"	- "I am getting a Syntax Error: unexpected EOF in my Python script below. Explain why it failed."

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

- A MAP function is a higher-order function that applies a specific transformation to every item in a collection and returns a new collection containing the results.
- Map function applies a given function to each element of an ~~item~~ Iterable and returns a map object, it processes and transform data efficiently

#example of Map()
list_of_strings = ['1', '2', '3', '4']
int_List = list(Map(int, List_of_strings))

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

- Cloud computing is the On-demand delivery of computing service
1) IaaS (Infrastructure as a service): Provides fundamental computing resources
example: Amazon EC2 (AWS)
2) PaaS (Platform as a service) - Provides a framework for developers to build, test and deploy Apps without managing underlying hardware.
example - Google App Engine
3) SaaS (Software as a service): Delivers ready-to-use software apps over the internet via subscriptions.
example: Google workspace.

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

- Version Control (VCS) is a software that tracks and manages changes to files over time.

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- Git is an open source Version Control system designed to track changes in source code, enable collaboration and maintain a complete history of project.
- Git is used by data analyst and developers because is the world's most popular distributed Version Control System. It is essential for data analyst and developers it tracks every change made, allowing users to revert to previous version if needed and facilitating collaboration with team.

* Commit - is a saved version of code representing a snapshot of the project's state at a specific point in time.

* Branch - A branch is a pointer to a commit, allowing you to create a separate copy of your projects where you can safely make changes without affecting the main code base.

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27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

- is An open-source web application used to create and share documents that contains live code and text and output.
- Python is the primary language used.
- it's popular: it allows data professionals to perform "literate programming, where they can run code in small cells and see their results immediately alongside their explanation

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

- Python is a high-level general-purpose programming language known for its readability and simplicity.

* Common Libraries: Pandas -> used for data manipulation and analysis.
* Numpy: Essential for numerical computing and handling large multi ~~media~~ dimensional arrays and matrices.
* Matplotlib: A foundation library for creating static, animated and interactive visualization like line graphs and histograms
* Scikit-learn -> The go-to library for machine learning, providing tools for data mining and predictive modeling

Section D: Data Ethics, Governance & Careers

Research Tip: *Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.*

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

- Data Privacy refers to the protection of personal information from unauthorized access or use.
- it is critical because it protects individuals from identity theft, discrimination and the misuse of their personal lives.
- GDPR is a comprehensive legal framework that sets strict rules on how companies collect and process the personal data of individuals
- it applies to all European Union and European Economic Area countries

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30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

- Data Ethics is a branch of ethics that studies moral problems related to data
- examples: Mass surveillance without consent: Collecting biometric or location data from individuals without their knowledge
* Problematic because it creates a feedback loop that unfairly target specific neighborhoods
- Predictive Policing bias: using historical crime data that reflects systemic projustice to deploy police
- Data Selling: problematic because it exploits personal information for profit

31. What is Data Governance? Define it and explain the key components of a good data governance framework.

- Data Governance is a structured approach that defines the processes, roles, policies and standards for managing an organization's critical data assets.
- Key comp: * DATA Standards -> Rules for data naming, formating and classification
- Data Quality -> Processes to ensure data is accurate, complete and reliable
- Policy and Compliance -> Ensuring data handling meets legal and internal requirements.

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.

- AI Bias occurs when algorithm produces results that are systematically prejudiced due to erroneous assumption in the machine learning process
- A Model is biased if its output unfairly disadvantages certain group of people based on traits like race, gender, or age.
- example: * Healthcare bias - Predictive diagnostic tools trained on ~~[illegible]~~ underrepresented data for women or minority groups may yield less accurate results.

33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

Role	Key differences	Skills	Tools
3 DATA Analyst	- focuses on data interpretation and business problem solving	- Statistical knowledge - Communication - Data Manipulation	- SQL, R - Python - Data Visualization software
2 DATA Scientist	- Applies advanced statistical and machine learning techniques to data analysis	- advanced statistical knowledge - Machine learning, Data Conditioning.	- R, Python - SQL, SAS - Data conditioning Tools.

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③ DATA Engineer	Building and Maintaining datapipeline/infrastructure	- Data warehousing - ETL, Database Mgmt.	- Spark, Hadoop, - SQL, - AWS/Azure
④ AI Engineer	- Developing and deploying AI Models and neural network	- Deep learning - NLP and - Soft ware ENG	- Pytorch, Keras - OpenAI, API C++

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

- BI refers to the procedural and technical infrastructure that collects, stores and analyzes the data produced by a company's activities
* BI Tools: Power BI - It allows users to connect to various data sources and create immersive, interactive reports.
* Tableau: known for its powerful data visualization capabilities. it helps businesses make data-driven decisions by providing a way to analyze data, spot patterns and make informed decisions based on the insights gained.

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

- Section A (DATA fundamentals)
- It explains how information is collected, stored and used to solve real problems. and I am currently building my knowledge on SQL, spreadsheet and basic data modeling as I am an admin. so I want to learn more about data quality, databases and how strong fundamentals support analytics and reporting.

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

- Cloud computing (Section C.
- I watched a video on youtube from Simplilearn page.
- I also consulted with my mentor about it

37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

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- In My Career these skills help me make better decision and stay relevant as more roles require data literacy. In everyday life, they help me understand and ~~see~~ use the tools around
- example: for my career, I plan to use data to track business ~~perfo~~ performance, spot trends and back up recommendations with evidence.

example: I can analyze sales reports to see which products perform best each month and suggest where to focus marketing

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

1. www.Geeksforgeeks.org
2. ~~google Copilot search~~ www.browserstack
3. www.guruqa.com/database-vs-datawarehouse
4. azure.Microsoft.com.
5. www.IBM.com/think/topics
6. www.IBM.com
7. Cloud.google.com
8. www.Oracle.com

Good luck! We believe in you.

Submit completed work to: rofhiwa@brightlighttutorials.co.za

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